DATA QUALITY AUDIT & RECONCILIATION REPORT

Project-AEGIS | DL3004 Data Warehousing & Business Intelligence | Phase 2 Artifact

***** MANDATORY ACADEMIC NOTICE: SYNTHETIC BENCHMARK DATASET (RANDOM SEED: 42) *****

This engineering report and its underlying data warehouse tables were compiled exclusively from a **synthetic test dataset** generated with a deterministic pseudo-random seed of **42** using 03_ETL/00_generate_synthetic_data.py. This data was engineered specifically to satisfy the **DL3004 DWBI university course grading rubric volume requirements** (≥100 business dimension rows, ≥10,000 fact rows) without exposing private federal credentials or executing live cloud API downloads.

Injected Quality Defect Total: 210 deliberate defects (whitespace variants, null keys, duplicates, sentinels, unparseable dates, and thermodynamic bounds violations) were injected into the Bronze landing zone to rigorously prove that the Silver cleaning and Gold validation pipelines catch and quarantine every single defect.

Strict Integrity Clause: The figures, facility names, and telemetry herein reflect synthetic benchmark generation and must never be cited as real historical federal operations from the EIA, EPA, or NOAA.

Medallion Layer	Lakehouse Schema	Dataset Record Count	Layer Description
Bronze Staging	dwbi_catalog.bronze_staging	**29,382**	Raw landed synthetic files (EIA-860, 923, 930, CAMPD, NOAA)
Silver Conformed	dwbi_catalog.silver_conformed	**29,222**	Conformed units, standardized text, coerced sentinels, deduplicated
Dead-Letter Quarantine	dwbi_catalog.quarantine	**30**	Rejected records with specific error reason codes
Gold Star Schema	dwbi_catalog.gold_warehouse	**16,182**	Kimball Star Schema facts (Emissions, Monthly Gen, Grid Demand)
Injected Defects Total	**synthetic_injection_log.csv**	**210**	**Deliberate defects 100% caught, cleaned or quarantined**

1. Data Quality Framework & Category Evaluation Summary

This audit report presents the automated validation results executed by the Data Quality Suite (03_ETL/05_data_quality_validation.py, Batch: TEST_SYNTH_INTEGRITY). A total of **66 individual checks** were executed across 6 core quality categories. Results: **65 PASS**, **1 WARN**, **0 FAIL**.

Check Category	Checks Evaluated	Pass Count	Warn Count	Fail Count	Category Health Status
COMPLETENESS	40	40	0	0	**PASS (100%)**
ACCURACY	4	3	1	0	**WARN (1)**
REFERENTIAL_INTEGRITY	8	8	0	0	**PASS (100%)**
UNIQUENESS	8	8	0	0	**PASS (100%)**
ROW_COUNT_RECONCILIATION	2	2	0	0	**PASS (100%)**
DATE_COVERAGE	4	4	0	0	**PASS (100%)**

Check Category	Checks Evaluated	Pass Count	Warn Count	Fail Count	Category Health Status
OVERALL LAKEHOUSE HEALTH	**66**	**65**	**1**	**0**	**HEALTHY (WITH WARN)**

2. Data Quality Category Threshold Specifications

Category	Quality Dimension	Formal Evaluation Threshold	Action on Violation
COMPLETENESS	Missing / Null Value Ratios	Null Ratio ≤ 5.0% on operational measures	Raise WARN; route unparseables to quarantine
ACCURACY	Cross-Agency Reconciliation	Discrepancy == 0.00 on measure conservation sums; 0.01 ≤ aux_ratio ≤ 0.16	Raise FAIL on sum mismatch; WARN on auxiliary ratio drift
REFERENTIAL INTEGRITY	Foreign Key Member Validity	Orphan Count == 0 (Hard Rule); Default Member (-1) Rate ≤ 5.0%	Raise FAIL on orphans (>0); WARN if default rate > 5.0%
UNIQUENESS	Primary & Composite Key Uniqueness	Duplicate Primary Keys == 0	Raise FAIL; deduplicate during Silver conformance
ROW RECONCILIATION	Conservation of Rows Across Layers	rows_in == rows_out + rows_quarantined + rows_removed	Raise FAIL on any unexplained row discrepancy
DATE COVERAGE	Calendar Span & Fact Referencing	1,461 Days (2021-2024); Unreferenced Fact Dates == 0	Raise FAIL if leap day missing or orphan date referenced

3. Deep-Dive Audit: Active WARN & FAIL Checks

Check ID	Table & Column	Metric Value	Threshold	Status	Engineering Root Cause & Plain-Language Explanation
ACC_AUX_LOAD_RECON	fact_monthly_plant_operations.net_generation_mwh	**0.0975**	0.01 <= aux_ratio <= 0.16	**WARN**	The fleet-average auxiliary load ratio is 0.0975 (9.75%), which sits within nominal bounds (1% to 16%). However, a small number of maintenance-outage plant-months report zero net generation while auxiliary load draws power, causing individual plant auxiliary load ratios to briefly spike outside the strict bounding envelope. This represents valid thermodynamic physical behavior during maintenance shutdowns.

4. Referential Integrity Audit: Foreign Key Default Member (-1) Rates

Referential integrity requires two distinct validation levels: (1) True orphan keys pointing to non-existent dimension records must be strictly 0; (2) Unmatched business keys safely resolving to the default member (-1) must not exceed operational thresholds (WARN > 5.0%, FAIL > 25.0%).

Target Fact Table	Foreign Key Column	Audit Check Metric	Metric Value	Evaluation Threshold	Audit Status
fact_hourly_emissions_weather	plant_key	-1 Default Rate	**0.0%**	WARN > 5.0%, FAIL > 25.0%	**PASS**
fact_hourly_emissions_weather	plant_key	Orphan FK Count	**0 orphans**	0	**PASS**
fact_hourly_emissions_weather	station_key	-1 Default Rate	**0.0%**	WARN > 5.0%, FAIL > 25.0%	**PASS**
fact_hourly_emissions_weather	station_key	Orphan FK Count	**0 orphans**	0	**PASS**
fact_monthly_plant_operations	plant_key	-1 Default Rate	**0.0%**	WARN > 5.0%, FAIL > 25.0%	**PASS**
fact_monthly_plant_operations	plant_key	Orphan FK Count	**0 orphans**	0	**PASS**
fact_regional_grid_demand	ba_key	-1 Default Rate	**0.0%**	WARN > 5.0%, FAIL > 25.0%	**PASS**
fact_regional_grid_demand	ba_key	Orphan FK Count	**0 orphans**	0	**PASS**

5. End-to-End Medallion Row Count Reconciliation

This reconciliation proves that data is conserved across the Medallion Lakehouse layers without unrecorded loss. Every single record in Bronze is either conformed to Silver, quarantined with an explicit error code, or purged as an identical duplicate.

Source Dataset	Bronze Raw Landed	Silver Conformed	Quarantined	Deduplicated	Row Conservation Balance
EIA-860 Generating Plants	129	125	4	0	129 = 125 + 4 + 0 (0 diff)
EIA-860 Generators Master (SCD-2)	380	375	0	5	380 = 375 + 0 + 5 (0 diff)
EIA-923 Monthly Generation & Fuel	1,688	1,680	8	0	1688 = 1680 + 8 + 0 (0 diff)
NOAA ISD Surface Weather Telemetry	12,540	12,540	0	0	12540 = 12540 + 0 + 0 (0 diff)
EPA CAMPD Hourly Flue Emissions	12,524	12,500	14	10	12524 = 12500 + 14 + 10 (0 diff)
EIA-930 Regional Grid Demand Telemetry	2,011	2,002	4	5	2011 = 2002 + 4 + 5 (0 diff)
TOTALS ACROSS ALL DATASETS	**29,272**	**29,222**	**30**	**20**	**CONSERVED (0 DISCREPANCY)**